MERN STACK

TRAINING TR-103 REPORT DAY 111

7 July 2026

Overview:

The eleventh day of the MERN Stack training focused on the Document Object Model (DOM) and different methods used to access HTML elements using JavaScript. We learned how the browser represents an HTML document as a tree structure and how JavaScript interacts with it to modify webpage content dynamically. We also practiced various DOM selection methods such as getElementById(), getElementsByClassName(), getElementsByTagName(), querySelector(), and querySelectorAll().

Learning Objectives:

- Understand the concept of the Document Object Model (DOM).
- Learn how JavaScript interacts with HTML using the DOM.
- Understand different methods to access HTML elements.
- Learn the use of getElementById().
- Explore getElementsByClassName() and getElementsByTagName().
- Learn the use of querySelector() and querySelectorAll().

Document Object Model (DOM)

The Document Object Model (DOM) is a programming interface that represents an HTML document as a tree structure. Every HTML element is treated as an object (node), allowing JavaScript to access, modify, add, delete, and style webpage elements dynamically without reloading the page.

Features:

- Represents an HTML document as a tree structure.
- Each HTML element is treated as an object (node).
- Allows JavaScript to update webpage content and styles dynamically.

Accessing Elements in DOM

JavaScript uses the document object to access HTML elements on a webpage.

1. getElementById()

The getElementById() method selects a single HTML element using its **id**.

Features:

- Selects one element.
- Uses the element's unique id.

Syntax :

```
document.getElementById("idName");
```

2. getElementsByClassName()

The getElementsByClassName() method returns all elements with the specified class name.

Features:

- Selects multiple elements.
- Uses the class name.

Syntax **:**

```
document.getElementsByClassName("className");
```

3. getElementsByTagName()

The getElementsByTagName() method selects all elements with a specific HTML tag.

Features:

- Selects elements by tag name.
- Returns all matching elements.

Syntax:

```
document.getElementsByTagName("tagName");
```

4. querySelector()

The querySelector() method returns the first element that matches a specified CSS selector.

Features:

- Selects the first matching element.
- Uses CSS selectors.

Syntax:

```
document.querySelector("CSS Selector");
```

5. querySelectorAll()

The querySelectorAll() method returns all elements that match a specified CSS selector.

Features:

- Selects multiple matching elements.
- Uses CSS selectors.
- Returns a collection of elements.

Syntax :

```
document.querySelectorAll("CSS Selector");
```

Conclusion:

By the end of Day 11, I developed a clear understanding of the Document Object Model (DOM) and learned how JavaScript interacts with HTML elements. I also practiced different DOM selection methods, enabling me to access and manipulate webpage content dynamically to create more interactive web applications.